Introduction to Multivariable Calculus (In Progress)

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Preface

1. Introduction

This text introduces a subject called *multivariable calculus*. This is typically the next logical step after a first course in calculus.

I first started writing this text for my own personal development, in order to teach the subject to myself; in that regard, it has been a valuable learning journey for me. My goal in this text is to mitigate some of the difficulties I encountered when first studying the subject, so that hopefully others can benefit from my experiences.

2. Prerequisites and Notation

This text has only one prerequisite of calculus, at the level of Art of Problem Solving's *Calculus* (in particular involving the delta-epsilon definition of limit.)

From my experience, most mathematically rigorous texts on this subject assume more prerequisites, and they require familiarity already with non-rigorous multivariable calculus. However, here we build up to a fully rigorous presentation that is accessible to anyone encountering this subject for the first time. All additional background is provided either directly in the text or in the appendix; in particular, the appendix includes treatments of linear algebra and three-dimensional coordinate geometry.

We also assume familiarity with standard and common mathematical notation. For example:

1. $\mathbb{R}$ denotes the set of real numbers, $\mathbb{Z}$ the set of integers, and $\mathbb{Q}$ the set of rational numbers.

2. For either of these three sets, attaching the subscript $+$ produces the subset of positive members, so for example $\mathbb{Z}_+$ is the set of positive integers.

3. Similarly, for either of these three sets, attaching the subscript 0 produces the subset of nonnegative members, so for example $\mathbb{R}_0$ is the set of nonnegative real numbers.

4. For any two sets A and B, $A \times B$ denotes the set of ordered pairs where the first "coordinate" is from A and the second is from B: $A \times B = \{(a, b) : a \in A, b \in B\}$. For example, if $A = \{1, 2\}$ and $B = \{1, 2, 3\}$, then $A \times B = \{(1, 2), (1, 3), (2, 2), (2, 3)\}$.

5. In particular, for any set S, $S^2 = S \times S$ denotes the set of ordered pairs of elements of S: $S^2 = \{(a, b) : a, b \in S\}$. For example, if $S = \mathbb{R}$, then $S^2 = \mathbb{R}^2$ is the set of ordered pairs of real numbers (which geometrically is the set of two-dimensional points, aka two-dimensional space.)

6. For any three sets A, B, and C, $A \times B \times C$ denotes the set of ordered triples where the first "coordinate" is from A, the second from B, and the third from C: $A \times B \times C = \{(a, b, c) : a \in A, b \in B, c \in C\}$.

7. In particular, for any set S, $S^3 = S \times S \times S$ denotes the set of ordered triples of elements of S: $S^3 = \{(a, b, c) : a, b, c \in S\}$. For example, if $S = \mathbb{R}$, then $S^3 = \mathbb{R}^3$ is the set of ordered triples of real numbers (which geometrically is the set of three-dimensional points, aka three-dimensional space.)

8. For any n sets $A_1, ..., A_n$, $A_1 \times ... \times A_n$ denotes the set of all ordered n-tuples where the ith "coordinate" is from A_i: $A_1 \times ... \times A_n = \{(a_1, ..., a_n) : a_i \in A_i\}$.

9. In particular, for any set S, S^n, or S $\times$'d with itself n times, is the set of ordered n-tuples of elements of S: $S^n = \{(a_1, ..., a_n) : a_i \in S\}$.

10. For $x \in A_1 \times ... \times A_n$, x_i denotes the ith "coordinate" of x, which would be

an element of A_i. For example, if $x = (4, 5, 6) \in \mathbb{R}^3$, then $x_1 = 4$, $x_2 = 5$, and $x_3 = 6$.

3. Differences From Other Texts

Some texts only cover multivariable calculus up to three dimensions, while others cover multivariable calculus in n dimensions from the get-go. We instead cover multivariable calculus up to three dimensions first, and afterwards we generalize some of our theory to n dimensions. This allows for a seamless transition from a typical intro course to the higher-dimensional calculus that will become more commonplace later on in math.

4. Practice Problems

If you want more practice problems in multivariable calculus, we recommend that, in addition to standard "plug-and-chug" exercises that may involve just applying formulas in obvious ways, you also look at "brainteaser" problems that involve more creative combinations of theorems. The Putnam Mathematical Competition is a famous college-level extracurricular contest that features these kinds of "brainteaser" problems for typical undergraduate math subjects, including multivariable calculus.

5. Feedback

I would appreciate any comments, feedback, or corrections. You can contact me at `https://riemannzeta5.com/contact`.

Now, without further ado, let's begin!

Part I.

Introduction

1. Primer: What is Multivariable Calculus?

At this point, you've probably finished a course in calculus, and you've seen that the next logical step is to take multivariable calculus. But what is this subject about?

Well, in calculus, we studied concepts like limits, derivatives, and integrals. These concepts were defined for single-variable functions, or functions with one real-number input and one real-number output. As such, what we studied is properly called *single-variable calculus*.

On the other hand, *multivariable calculus* is the extension of those concepts to multivariable functions, or functions with potentially multiple real-number inputs or outputs. Examples include the functions $f(x, y) = 2x + y$ (multiple inputs), $f(x) = (x, 2x)$ (multiple outputs), and $f(x, y) = (2x, x + y)$ (multiple inputs and outputs.) We can represent multiple numbers as an element of $\mathbb{R}^n$ for $n > 1$, thus we can also say that multivariable calculus concerns functions $f : \mathbb{R}^n \to \mathbb{R}^m$ where at least one of n and m is greater than 1.

Given the greater number of situations which can be modelled when we allow more data attributes to serve as inputs or outputs, it is not too surprising that multivariable calculus has a plethora of applications across disciplines in applied math and the sciences. Indeed, this subject is a fundamental tool of physics, economics, statistics, and many other fields.

Now, it turns out that it is really hard, if not impossible, to visualize calculus concepts for functions where n or m is greater than 3. Since calculus is inherently motivated by geometry, visualization is important to motivate and define the

concepts we work with. There does turn out to be a calculus of functions where n or m can be greater than 3, but first we will cover the theory for $n, m \leq 3$, which in fact is the material of a typical first course in multivariable calculus. Afterwards, we will generalize some of our theory to work with all n, m. This is a great stepping-stone to some later math subjects like manifold calculus, which frequently work with general n, m. (And those subjects end up having further applications, even beyond math to fields like physics.)

When I first studied multivariable calculus, there were a few major challenges I encountered, especially compared to other math subjects. To prepare you for them, I'll discuss them now before we jump into the math, and I'll offer some tips on mitigating them.

First, there are many new concepts in multivariable calculus compared to single-variable calculus. It turns out that there are many different kinds of derivatives and integrals we can define and study for multivariable functions, unlike single-variable calculus where there is (for the most part) just one kind of derivative and integral. The diversity of these concepts is one of the reasons why multivariable calculus is such a rich and exciting subject. However, it can also make things more confusing and overwhelming, especially on a first encounter.

To address this, it can be helpful to be prepared for, and to expect, this "concept explosion." It is also useful to clarify which derivative or integral is being used, which can sometimes be hard when you are self-studying and trying to piece together various resources – more on that next. In this text, we will try to alleviate this by always being clear about which concept we are concerned with at a given time, and by also spending more time covering a particular derivative or integral on its own.

Also, if you are a self-studier, multivariable calculus can be particularly frustrating to figure out. At least in my experience, most of the introductory resources were not rigorous, and when the material was finally covered rigorously later on, it was heavily generalized to settings requiring significantly more prerequisites. I tried to find or piece together proofs that did not use these prerequisites, for introductory-scope special cases of the highly generalized theorems. However, when such proofs were taken from different sources, it was hard to avoid cir-

cularity (a proof of theorem A uses theorem B, but then a proof of theorem B found somewhere else uses theorem A.)

This text came about after figuring all that out, and thus it fully avoids circularity. However, in general, if you want to self-study a concept in multivariable calculus, it is helpful to use a resource that features a fully developed curriculum, as opposed to a one-off exposition on that particular topic.

Multivariable calculus is ultimately an exciting subject, so don't be too discouraged by some of these difficulties. At the end of the day, it'll be incredibly rewarding once you study and understand this material. So with that in mind, let's move forward!

Part II.

Functions With Two or Three Outputs

2. Vector Calculus

2.1. Vector Limits

2.2. Vector Derivatives and Integrals

Part III.

Functions With Two Inputs

3. Two-Dimensional Limits

3.1. Definitions and Properties

Section to be added.

3.2. Continuity and Surfaces

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4. Two-Dimensional Derivatives

4.1. Partial Derivative

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4.2. Directional Derivative

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4.3. Differentiability

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4.4. Multivariable Chain Rule

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6.2. Surface Integral

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Part IV.

Functions With Three Inputs

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7.1. Scalar Fields

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7.2. Vector Fields and the Del Operator

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8. Volume and Triple Integrals

8.1. Definitions and Basic Properties

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8.2. Analogues of Previous Theorems

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Part V.

Calculus on Boundaries

9. Boundaries, Orientation, and Topology Background

9.1. Boundary, Interior, and Exterior

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9.2. Open and Closed Sets

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9.3. Connected Sets

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9.5. Orientation

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10. The Fundamental Theorems

10.1. The Gradient Theorem

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10.2. Gauss' Theorem

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10.3. Green's Theorem

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10.4. Stokes' Theorem

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Part VI.

Functions With Any Number of Inputs and Outputs

11. Higher-Dimensional Geometry and Linear Algebra

11.1. Introduction and Philosophy

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11.2. Points and Function Graphs

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11.3. Lines and Hyperplanes

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Special Topics

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14.1. Polar Coordinates

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14.2. Cylindrical Coordinates

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14.3. Spherical Coordinates

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15. Additional Theorems of Multivariable Calculus

15.1. Converse of the Gradient Theorem

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15.2. Helmholtz Decomposition

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Part VIII.

Appendix

A. Three-Dimensional Coordinate Geometry

A.1. Introduction and Function Graphs

Section to be added.

A.2. Lines, Planes, and Rectangular Solids

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A.3. Cylinders and Spheres

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